

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Built full-stack knowledge-graph app with dual-DB persistence (PostgreSQL + Neo4j), enabling data interoperability across stores; offloaded recursive queries to Neo4j, cutting p95 48% (850ms → 440ms) and enabling scalable multi-hop traversals.
- Refactored Cypher with relationship-type indexing; traversals 3× faster than SQL self-joins, removing full-table scans.
- Integrated dual-provider OAuth 2.0 (Google + GitHub) with httpOnly JWT, refresh rotation, and user-scoped access for secure API integration and cross-system interoperability.
- Implemented GitLab CI/CD (build, test, containerized deploy) with branch-protected merges and 85% coverage across 35+ tests for production-ready releases.

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding via GitHub API + Slack SDK integration, granting repo and workspace access in 30s and cutting setup time 97% across 100+ members.
- Replaced paper sign-in with QR-driven check-in (AWS S3 token caching), enabling real-time tracking and cutting processing time 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with 90% coverage gates across 20+ tests.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered multi-client agentic voice assistant that resolves implicit intent and dispatches real tasks across concurrent voice streams.
- Reduced API response time 73% by offloading RAG pipeline to scalable async Redis/Celery task queue.
- Built Electron + Next.js client with embedded HTTP server for OAuth callbacks; IPC and webhook server deliver sub-50ms event propagation.
- Deployed FastAPI on AWS Lambda with lazy-loaded ML models, reducing cold-start 86% (6,643ms → 930ms).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow* Feb. 2025

- Won Interactive Media Track: gesture-controlled wizard duel (full-body spells, block, abilities vs. AI opponent).
- Built MediaPipe Pose pipeline mapping 33 skeletal landmarks to game actions; sustained 60 FPS with sub-20ms input lag.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash
Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis
Databases PostgreSQL, Neo4j, MongoDB
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitLab CI/CD, CI/CD Pipelines
Standards HL7, FHIR, EHR (Epic, Cerner, Meditech)